Oriented Ship Detection Based on Soft Thresholding and Context Information in SAR Images of Complex Scenes

Chuan Zhang, Gui Gao, *Member, IEEE*, Jia Liu, and Dingfeng Duan

***Abstract*— The detection of ships encompasses an abundance of applications within the domains of fishery management, marine rescue operations, and maritime monitoring. In recent years, a multitude of detectors based on deep learning have been utilized for the purpose of ship detection using synthetic aperture radar (SAR) images. However, disturbed by the strong scattering background on land and the influence of the SAR target scale, the existing detectors face great challenges in detecting inshore small ships. To solve this problem, this article proposes an oriented ship detection network for SAR images based on soft threshold and context information. First, a soft-threshold quantization module (STQM) based on the soft threshold function is designed to alleviate the interference of background noise on the feature map. Second, a local and global context fusion module (LGCFM) is designed to capture the contextual information of the target to enhance the detection of small targets. Third, the inclusion of the center loss in the loss function serves to impose additional constraints on the center coordinates and shape of the oriented bounding box. This is done to achieve a more balanced distribution of loss contributions across the various variables and to mitigate the target's susceptibility to variations in the shape of the ground-truth bounding box. Finally, the proposed network is tested on the publicly available mini-rotated-high-resolution SAR images dataset (MR-HRSID) and Rotated SAR Ship Detection Dataset (R-SSDD) datasets. The results from experiments demonstrate that our approach attains state-of-the-art detection capabilities for inshore ships and small targets, while also effectively mitigating interference from terrestrial noises.**

***Index Terms*— Context information, oriented ship detection, soft thresholding, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR) possesses several advantageous characteristics, including its ability to operate continuously during the day and under various weather conditions. Additionally, SAR enables wide-area mapping and offers high-resolution imaging capabilities. These attributes contribute significantly to SAR's crucial role in the detection and surveillance of marine targets within the domain of Earth observation [1]. Ship detection is a key application in SAR imagery, including two subtasks of localization and classification, which are important in both military and civilian fields. Accurate location information provides a critical information base for the government to quickly respond to maritime emergencies [2]. According to different detection paradigms, we can roughly categorize SAR image target detection into two types: traditional threshold-based detection algorithms and deep learning-based methods.

Among the traditional threshold-based detection ship algorithms, the most used is the constant false alarm ratio (CFAR) algorithm. Such algorithms typically perform statistical modeling and threshold estimation of clutter under the assumption of a constant false alarm rate and then use the threshold to filter out ships [3], [4]. For the statistical model of clutter, the common ones are Weibull [6], K [7], generalized K [8] distributions, and Gaussian distribution. The utilization of these distribution models is prevalent in characterizing the statistical characteristics of marine clutter. For example, Ai et al. [9] proposed a new correlation algorithm using the strong correlation of ship grayscale and implemented a joint CFAR detection based on correlation. Leng et al. [10] proposed a bilateral CFAR algorithm for ship detection by combining the intensity distribution and spatial distribution in SAR images. However, conventional ship detectors rely heavily on ocean clutter modeling. Modeling offshore scenes is a challenging task due to the complexity of real-world application scenarios. As a result, the traditional algorithms perform better in the open ocean and the detection performance drops drastically when detecting inshore ships with complex backgrounds.

In recent years, convolution neural network (CNN) has been widely employed in computer vision with its excellent feature extraction capabilities and has shown high robustness and efficiency in numerous vision applications.

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CNN-based target detection techniques are typically divided into two classes based on distinct detection head structures: single-stage target detection networks (represented by SSD [11]) and two-stage target detection networks [represented by Faster region-based CNN (R-CNN)]. Both types of target detection networks and their derivatives have fetched excellent detection results in the field of target detection. Currently, researchers working in the field of SAR are also interested in CNN-based target detection methods. With mature SAR imaging techniques providing researchers with a large amount of available data, several SAR datasets suitable for deep learning have been released, such as the HRSID dataset [12] and the SSDD dataset [13]. Based on these publicly available datasets, CNN-based ship detection algorithms for SAR images have been proposed one after another. For example, to achieve fast target detection with high accuracy, Zhu et al. [14] utilized spatial pyramid pooling and mosaic data augmentation to construct the high-speed and high-accurate detector (H^2Det). Ai et al. [15] integrated truncated statistical CFAR detector, CNN, and support vector machine three algorithms and proposed a multiscale rotation invariant Haar-like (MSRI-HL) feature-based integrated convolutional neural network. To fulfill the timeliness of ship detection in SAR images of big sceneries, Miao et al. [16] suggested a lightweight framework based on a threshold neural network. Sun et al. [17] proposed a high-resolution SAR image ship detection method based on an anchor-free strategy. In addition, they also proposed a You Only Look Once (YOLO)-based arbitrary direction SAR ship detector using bidirectional feature fusion and angle classification [18]. Zhang et al. [19] constructed a high-precision and high-speed SAR ship detection network HyperLi-Net using multiple external modules. Although these CNN-based ship detection algorithms have achieved good performance on publicly available datasets, CNN target detection algorithms based on horizontal–vertical boxes label strategies introduce more background clutter, which causes difficulties in neural network learning and reduces the detection accuracy.

In summary, although the SAR imaging system has unique advantages in maritime target detection, the detection performance of inshore ships is still limited by the interference of land background and other factors. First, the scattering intensity of complex land backgrounds such as buildings and harbors in the imaging scene is like that of ships [20], which brings a great deal of disruption to the ship detection algorithm and may lead to more false alarms in inshore ship detection. Second, because of the complexity and variability of the marine environment, the clutter in high-sea state scenarios can seriously affect the imaging quality of SAR images [12], [21], making the ship less visible in the image and leading to missed ship detection. Most importantly, due to differences in the viewing angle of the SAR imaging system, ships may appear to be oriented arbitrarily in the SAR image [13]. The classical horizontal box labeling strategy is unable to fully fit the position of the ship, resulting in a low accuracy of the ship's positioning (see Fig. 1). To address this issue, it is important to precisely determine the ship's orientation angle and conduct the detection of the orientated enclosing boxes.

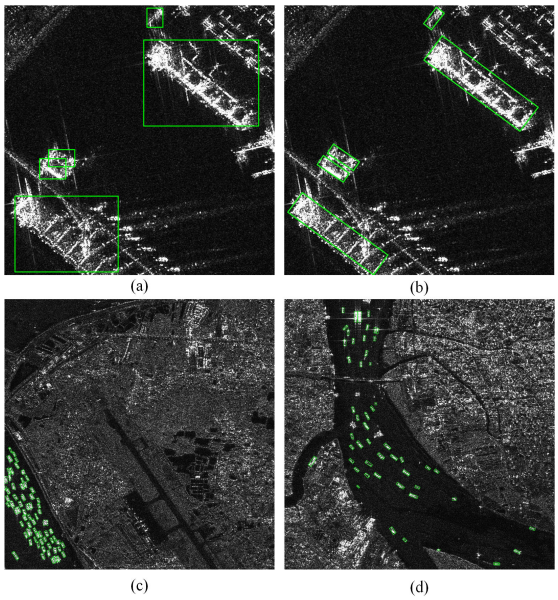


Fig. 1. Samples from the HRSID dataset: (a) denotes targets labeled with horizontal–vertical boxes; (b) denotes targets labeled with rotated boxes; and (c) and (d) denote ships in inland waterways.

A. Problem and Motivation

In SAR image target detection, CNN-based target detection algorithms have made significant breakthroughs after several years of development. However, the inherent defects of SAR images limit the performance of SAR image target detection. The specifics are as follows.

1) Strategies based on horizontal–vertical boxes make it difficult to localize the positional accuracy of ships. Due to the arbitrary angle of the ship's heading on the sea surface, the labeling technique that relies on a horizontal–vertical box will unavoidably include unnecessary clutter in the backdrop. This not only diminishes the accuracy of the ship's positioning but also affects subsequent postprocessing procedures [22], [23].
2) The task of detecting ships in inshore environments poses significant challenges. Due to the complex architectural composition of land-based structures, it is common for extraneous components, such as buildings situated inland, to exhibit pronounced scattering effects. This phenomenon has a resemblance to the imaging outcomes observed in the context of ship imaging near the coast. The presence of land backgrounds often leads to the occurrence of several false alarms in the detection outcomes [24], [25].
3) The task of detecting small-scale ships in SAR images is extremely difficult. In SAR images, some small-scale ships occupy only a dozen pixels on the SAR image, and their information is very small. After several downsampling operations in the neural network, the small-scale ships may disappear from the feature map, which leads to the missed detection of small-scale targets [26].

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B. Contribution

In this article, we propose an anchor-based oriented ship detection network for SAR images. In this network, we design a soft-threshold quantization module (STQM) for filtering background noise in the scene and a local and global context fusion module (LGCFM) for improving the detection performance of small targets. Overall, the contributions of this article are as follows.

1) The STQM is designed to filter background noise in complex scenes. In the inshore scenario, to filter the land noise while preserving the ship target, the STQM is designed to enable the network to adaptively learn the threshold parameter to effectively filter the noise in the feature mapping.
2) The LGCFM is proposed to retain more shallow features to improve the detection performance of small targets. To improve the detection performance of small targets, the LGCFM is constructed for the fusion of shallow detail features (e.g., texture and edges) and deep semantic features. The shallow detailed features are used to improve the ship localization accuracy and the deep semantic features are used to improve the ship classification accuracy.
3) An oriented ship detection network for complex scenes of SAR images is proposed, and the mini-rotated-HRSID (MR-HRSID) dataset is labeled based on the horizontal–vertical boxes of the HRSID dataset. Experiments are conducted on R-SSDD and MR-HRSID to verify the effectiveness of the proposed network.

The remaining sections are organized as follows. Section II details the oriented ship detection proposed network. Section III describes the experiments of this article. Section IV summarizes this article.

II. Methodology

In this section, the network architecture proposed in this article is described in detail, which is shown in Fig. 2. The network first extracts the multilevel features C_3, C_4, and C_5 from the input SAR image through the backbone network. Then, the feature map C_3 is input into the STQM to adaptively filter the noise to get C_3'. Next, the feature maps C_3', C_4, and C_5 are fed into the feature pyramid network (FPN) [29] to get P_3, P_4, P_5, P_6, and P_7, and the feature map C_3' is also input into the LGCFM to get C_3''. After that the feature maps P_3 and C_3'' are fused in a concatenated manner to get P_3'. Finally, the feature maps P_3', P_4, P_5, P_6, and P_7 are input into the detection head for prediction. In Sections II-B and II-C, we will focus on STQM and LGCFM.

A. Backbone and FPN

For a CNN-based target detection network, the feature extraction network backbone affects the final detection performance. After more than a decade of development, several backbones have been developed in the field of computer vision target detection, such as VGGNet [27], ResNet [28], and so on. Among them, ResNet performs the best and is widely used in tasks such as target detection, instance segmentation, and object classification, due to its powerful feature extraction capability and excellent task adaptation. Therefore, in the proposed network, one of the versions of ResNet (ResNet50) is selected as the backbone. ResNet50 performs downsampling operations on the input data with steps 4, 8, 16, and 32 to obtain the feature maps, respectively. Then, feature maps are input into the FPN to perform feature fusion. The FPN is a top-down feature fusion module that can integrate semantic information from deep features into shallow features, which helps multiscale target detection. Finally, the features output from FPN are regressed and classified.

B. STQM

In SAR images, pixels that are not related to the target are called noise pixels. As shown in Fig. 1, buildings on land are like ships in terms of scattering intensity, shape, and contour, and these noises can greatly reduce the detection performance of the algorithm. Conventional ship detection algorithms for SAR images usually detect ships from clutter after constructing a ship detector by using some probability distributions for threshold estimation. For example, Zhang et al. [30] proposed a ship detection algorithm for SAR images based on a completely polarized covariance difference matrix. However, the threshold estimation of such algorithms usually requires a priori knowledge from experts and previous land and sea segmentation. Once the scene contains a land background, the threshold estimation is biased, leading to many false alarms in the detection results.

For the past 20 years, a soft threshold has been a crucial component of signal-denoising techniques [31], [32]. In general, the original signal is first transformed to a near-zero digitally unimportant domain, and then, the near-zero features are converted to zero by soft thresholding. A typical example is wavelet thresholding, which is a classical signal denoising method that includes three steps: wavelet decomposition, soft thresholding, and wavelet reconstruction. The approach to wavelet thresholding is to create a filter that transforms helpful information into positive or negative features while transforming noisy information into features close to zero. Such a filter design, however, requires substantial signal processing knowledge and has long been a difficult task. The availability of CNNs provides a new way of thinking. Instead of requiring an expert to perform the manual design of the filter, the CNN automatically learns the filter through a gradient descent algorithm. Therefore, to filter the background noise in the feature map and reduce the interference of the terrestrial background on the neural network, based on the soft threshold algorithm, the STQM is designed to eliminate the noise contained in the feature map. The soft threshold function is defined as follows:

$$y = \begin{cases} x - \tau & x > \tau \\ 0 & -\tau \le x \le \tau \\ x + \tau & x < -\tau \end{cases} \tag{1}$$

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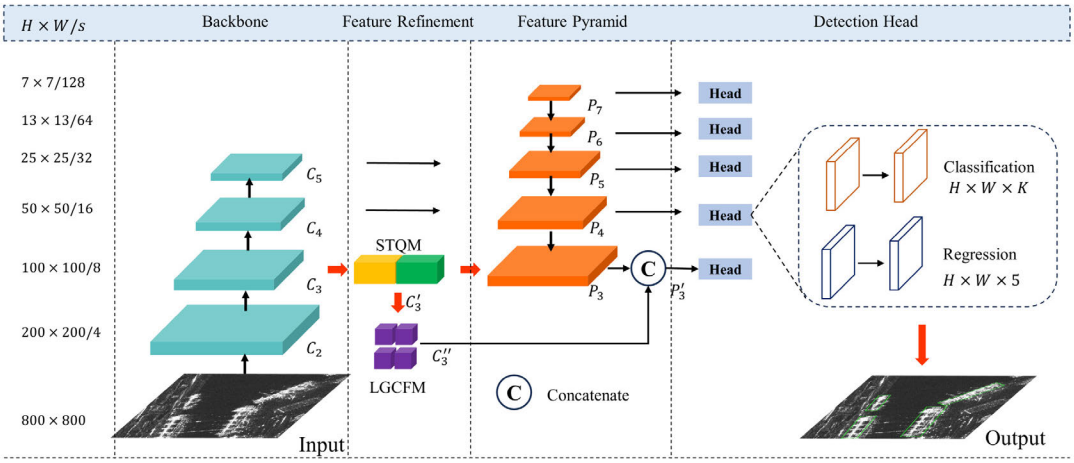


Fig. 2. Proposed network structure.

where x denotes the input signal, y denotes the output signal, and τ denotes the threshold. Its derivative can be expressed as

$$\frac{\partial y}{\partial x} = \begin{cases} 1 & x > \tau \\ 0 & -\tau \le x \le \tau \\ 1 & x < -\tau. \end{cases} \tag{2}$$

From (2), the soft-threshold function is derivable, and the output value of its derivative is either 0 or 1 and thus does not cause the gradient of the neurons in the network to vanish or explode.

Algorithm 1 Soft-Threshold Quantization Module

Input: Feature map C_3
Output: Feature map C_3'

1: $x_{abs} = abs(C_3)$
2: $x = GlobalAdaptivaPool2d(x)$
3: $x' = Flatten(x)$
4: $z = FC(x)$
5: $\alpha = Sigmoid(z)$
6: $\tau = x' \otimes \alpha$
7: $T = x_{abs} - \tau$
8: $G = max(T, 0)$
9: $C_3' = sign(C_3) \otimes G$

Based on the above threshold function and combined with the low-dimensional feature C_3, the STQM is designed, and its structure is shown in Fig. 3 and Algorithm 1. First, the C_3 obtains x by the absolute operation and 2-D global average pooling (GAP). Second, the feature x is input into the fully connected layer (including Linear, BatchNormal, and Rectified Linear Unit (ReLU) operation) and Sigmoid activation function, and the Flatten function to get α and x', respectively; third, α and x' are subjected to elementwise multiplication to get the final output C_3'. The motivation for designing STQM in this way is that, on the one hand, data in the real world, to a greater or lesser extent, contain some redundant information. STQM is embedded in the network to eliminate redundant information. On the other hand, the amount of redundant information in each sample is often different. Therefore, the attention mechanism is used to

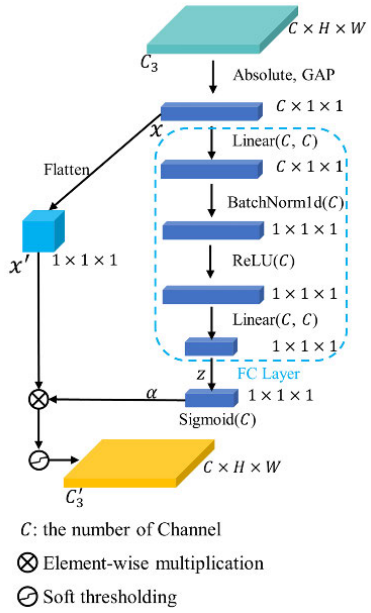


Fig. 3. STQM structure.

adaptively set different thresholds for each sample according to the situation of each sample. In addition, the reason we apply the STQM to C_3 is that small targets in low-dimensional features usually have large gradient values, but they are also accompanied by a lot of noise. Only by applying the STQM to low-dimensional features, can we filter out the noise and retain more detailed information about the small targets. In addition, high-dimensional features contain more semantic information, which helps to classify the targets. If we apply the STQM to high-dimensional features, it will filter out the semantic information learned by the network, which will lead to a decrease in the accuracy of the network's classification.

C. LGCFM

Humans can recognize scenes or objects in the field of view using relationships between people, environments, and objects [33], [34]. This captured semantic or spatial correlation

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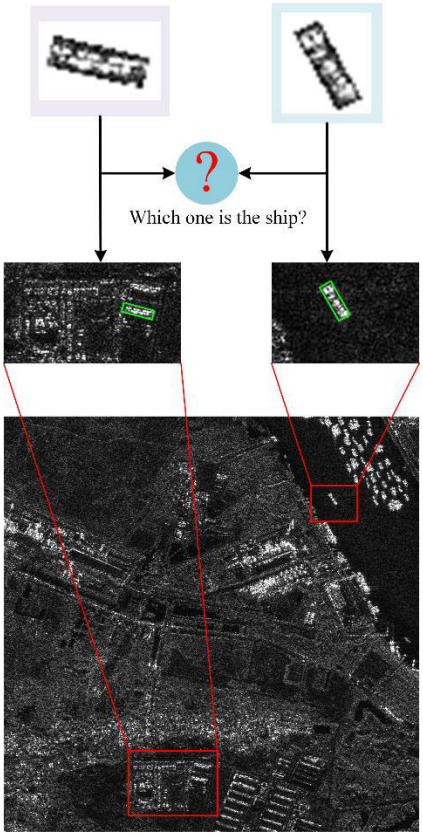


Fig. 4. Contextual information detailing.

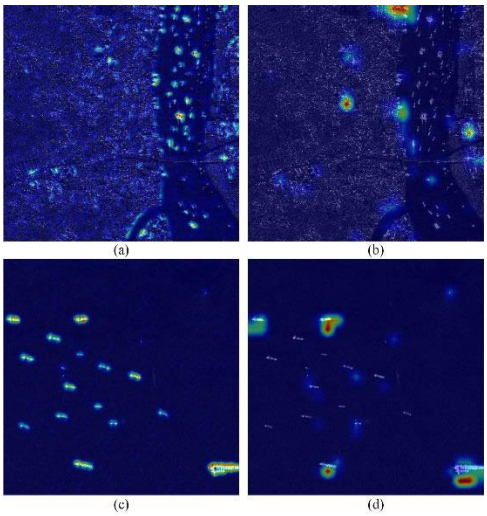


Fig. 5. Heat maps in the neural network: (a) and (c) denote the heat map corresponding to the feature map C_3 and (b) and (d) denote the heat map corresponding to the feature map C_5.

information we call contextual information, such that a priori knowledge conveys information beyond the evidence or cues intended to be expressed in the target area. Contextual information is not only crucial in the human visual system [35], but also plays a key role in tasks of scene understanding, e.g., in target recognition [36], semantic segmentation [37], and instance segmentation [38]. Interestingly, contextual information can sometime provide more important decision support than the target itself, especially in environments with poor visual quality [34], [35]. For example, in SAR images, we rely on two main aspects to discriminate whether an object is a ship or not: 1) the scattering intensity of the target and 2) the location of the target. Only when an object's scattering intensity is stronger than that of its surroundings and it is at the sea surface or docked at the shore, this object is likely to be considered as a ship. As shown in Fig. 4, the left bright region is located on land, and the right bright region is located on the ocean; combining the location of these two regions, we can easily distinguish which region is the real ship. However, if we only rely on the scattering intensity of the region, the ship cannot correctly be determined. The reason is that buildings on land have complex structures that produce scattering intensities like those of ships. In other words, in SAR images, the background also provides us with a basis for determining the target.

In addition to providing a basis for target decision-making, contextual information can effectively improve the detection of small targets. For instance, Hu and Ramanan [39] studied how to efficiently encode regions outside the object range and represent local contextual information in a scale-invariant manner to recognize tiny faces in the field of computer vision. PyramidBox makes the most of contextual information to find tiny, blurry faces that blend in with the surroundings [34]. A context-aware region of interest (RoI) pooling layer [40] is introduced by SINet [41] to preserve contextual information if the original RoI pooling operation breaks the structure of small targets. To enhance the effectiveness of small objects, Bell et al. [42] captured the correlation between an object and the overall scene (global context) as well as the correlation between an object and its nearby instances (local context). In SAR images, especially within inland waterways, many small-scale ships are included. These small-scale ships contain less information because of the small image area they occupy in the image, and as the depth of the neural network continues to deepen, the information contained in such objects will gradually decrease. As shown in Fig. 5(a) and (c), in the low-dimensional feature map C_3, the location where the small target is located has a high gradient value both in the inland riverway and in the open sea, and its contour and edge features are clearer, which is suitable for the localization and detection of small targets [35]. In the high-dimensional feature map C_5, its low resolution mainly extracts the semantic features of the target, resulting in small gradient values for small targets, and only some targets with larger scales show larger gradients. Therefore, high-dimensional features such as C_4 and C_5 are not suitable for the detection of small targets, especially in SAR images.

Based on the above analysis and combined with the actual situation of SAR images, the LGCFM is constructed, as shown in Fig. 6. In LGCFM, the feature map C_3' is first input into the average pooling layer with kernel sizes of 1×1, 2×2, 3×3, and 6×6, respectively, then upsample the output feature map by using the bilinear interpolation and concatenate to obtain the feature map C_3'', and finally fused again with the FPN output P_3 to obtain P_3'. The motivation for fusing P_3 and C_3'' is that,

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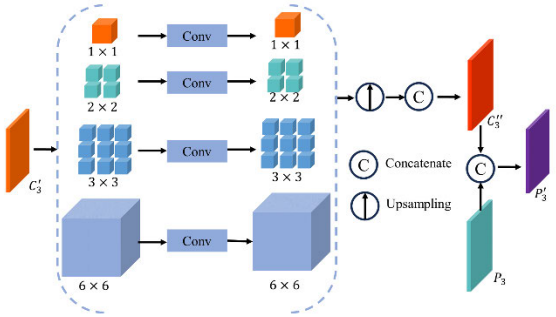


Fig. 6. LGCFM.

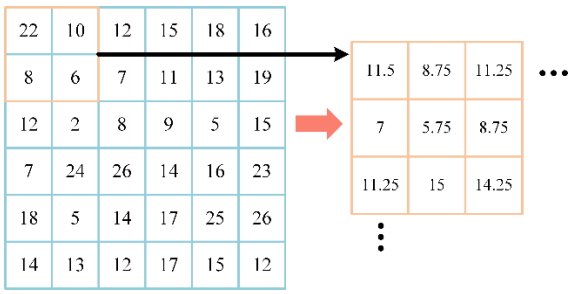


Fig. 7. Average pooling flow with a kernel size of 2 and stride of 1.

first, feature figure P_3 is obtained after top-down multiscale feature fusion. As a result, high-level semantic information is included in P3, which helps with target classification. Second, the feature map C_3'' obtained after LGCFM processing contains enhanced shallow detailed information, which is helpful for ship positioning. Therefore, P_3 and C_3'' are fused to get P_3', so that P_3' contains both semantic features and shallow detailed information, which improves the ability of feature map P_3' to classify and locate small targets.

In this module, the convolution kernel sizes [1, 2, 3, 6] are used. When kernel size = 1, 2, and 3, CNN can learn local features between the target and the target's surroundings. When kernel size = 6, the receptive field of the convolutional kernel is enlarged, and the CNN can learn the coupling relationship between the target and a large range of environments. That is, the global relationship between the target and the environment is learned. As mentioned earlier, in environments with poor visual quality, local and global information (contextual information) can provide better decision support than the ship itself. Therefore, in the inland channel of the SAR image (which contains a large number of small targets, and its visual quality is poor), the classification accuracy and positioning accuracy of the ship can be effectively improved by using the local and global information of the ship.

In LGCFM, the equations for the average pooling layer are as follows:

$$F_{\text{out}} = \text{AvgPool2d}(k \times k, F_{\text{in}}) \tag{3}$$

where F_{out} denotes the pooling layer output features, F_{in} denotes the input features, and k denotes the kernel size, $k = 1, 2, 3, 6$. More specifically, as shown in Fig. 7, in the average pooling with the kernel of 2 and stride of 1, a single pixel point on the output feature map can see the positions of 4-pixel points on the input feature map. In other words, the receptive field of the pixel points on the output feature map is enlarged so that a single pixel point can learn more contextual information, which helps in the judgment of the target.

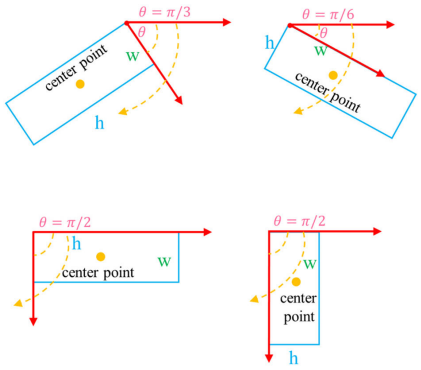


Fig. 8. Way of defining a rectangle in any orientation.

D. Loss Function

In this article, the multitask loss is used to optimize the final prediction network. Usually, the multitasking loss involves both classification and regression losses, whose expressions are as follows:

$$L(p_i, t_i) = \frac{1}{N_{\text{cls}}} \sum_i L_{\text{cls}}(p_i, p_i^*) + \lambda \frac{1}{N_{\text{reg}}} \sum_i p_i^* L_{\text{reg}}(\Delta t) \tag{4}$$

where λ denotes the weight factor. $p_i = \pm 1$ and $p_i^* \in [0, 1]$ denote the prediction confidence and true class label, respectively; and Δt denotes the difference between the predicted oriented box coordinates and the true oriented box coordinates. N_{cls} and N_{reg} denote the batch size during the training process; L_{cls} denotes the classification loss; and L_{reg} denotes the regression loss. In this article, focal loss is used to compute the classification loss [43] with the following expression:

$$L_{\text{cls}} = \begin{cases} -\alpha(1 - p_i^*)^{\gamma} \log(p_i^*), & \text{if } y = 1 \\ -(p_i^*)^{\gamma} \log(1 - p_i^*) & \text{if otherwise} \end{cases} \tag{5}$$

where $\gamma > 0$ is the adjustable factor and α is the weight factor.

In the regression stage, the five parameters $(x, y, w, h, \text{and } \theta)$ are utilized to define the rectangle in any direction. In the range of $(0, \pi/2]$, θ is defined as the acute angle of the rectangle side w for the x-axis, as shown in Fig. 8. This definition is consistent with that of OpenCV. Given the true value of the target $t_i = (x_i, y_i, w_i, h_i, \theta_i)$ and the predicted value of $t_i^* = (x_i^*, y_i^*, w_i^*, h_i^*, \theta_i^*)$, there are

$$\Delta t = t_i - t_i^*. \tag{6}$$

Thus, the regression loss [44] is expressed as

$$L_{\text{reg}}(\Delta t) = \text{smooth}_{L1}(\Delta t) \tag{7}$$

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where

$$\text{smooth}_{L1}(\Delta t) = \begin{cases} 0.5\Delta t^2, & \text{if } |\Delta t| < 1 \\ |\Delta t| - 0.5, & \text{otherwise.} \end{cases} \tag{8}$$

To balance the loss contributions among various variables, we also constrain the center coordinates and shape of the oriented bounding box with the center loss [45]

$$L_c = \frac{||c_g - c_a||_2}{0.5 \cdot ||d||_2} \tag{9}$$

where c_g and c_a denote the coordinates of the center point of the real frame and anchor, respectively. $d = (w, h)$ denotes the length and width of the ground-truth (GT) label, respectively. With this method, L_c is normalized to [0, 1]. In regression loss calculations, small-scale ships are more susceptible to inaccurate centroid predictions and smoothed L_1 loss. The normalized centroid loss makes small-scale ships less sensitive to the shape of the GT box, thus facilitating the detection performance of small targets.

Therefore, summarizing the above analysis, the loss function used in the proposed network is as follows:

$$\text{Loss}_{\text{all}} = L_{\text{cls}} + L_{\text{reg}} + L_c \tag{10}$$

where L_{cls} denotes the classification loss, L_{reg} denotes the regression loss, and L_c denotes the center loss.

III. Experiments

A. Datasets

On the HRSID [12] and R-SSDD [13] datasets, the proposed network is validated. The RadarSat-2, TerraSAR-X, and Sentinel-1 satellites, which have four polarization modes, HH, HV, VV, and VH, were used to construct SSDD. Including multiscale ships and numerous scenes in various surroundings, R-SSDD has a total of 1160 SAR images with resolutions ranging from 1 to 15 m (928 images of maritime scenes and 232 images of inshore scenes). These images contain 2456 ship targets, with 2.12 ship targets on average per image. The HRSID dataset contains 136 large scene SAR images with a resolution of 1–5 m, and the sliced scene contains 5604 images totaling 16 951 ships. However, the HRSID dataset is a horizontal–vertical box labeled and there are no rotated boxes that can be used directly. Therefore, to be able to test the HRSID dataset, the HRSID dataset is relabeled in the form of rotated boxes based on the horizontal–vertical box labels. In the HRSID dataset, there are 662 inshore samples and 4942 offshore samples, which are highly unbalanced. Therefore, to balance the number of inshore and offshore samples and to take into account the limited computer resources, among the new rotated-box-labeled samples, 534 inshore samples and 770 offshore samples are chosen, totaling 1304 samples, as the training set; 128 inshore samples and 272 offshore samples, totaling 400 samples, are used as the test set; and the whole rotated-box-labeled HRSID dataset totals 1704 samples, which is named MR-HRSID. In MR-HRSID, the inshore scenarios in the training and test sets cover the inshore scenarios in the entire HRSID dataset, so the difficulty of detection is comparable to that of the original dataset.

B. Evaluation Criteria

In this article, the average precision (AP) and average recall (AR) are used to evaluate the detection performance of the proposed network. Given an intersection over union (IoU) threshold of 0.5, AP is defined as follows:

$$\text{AP} = \int_0^1 P(R)\text{d}R \tag{11}$$

where P denotes the precision and R denotes the recall. Precision and recall are defined as follows:

$$P = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}}, \quad R = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{12}$$

where N_{TP}, N_{FP}, and N_{FN} denote the positive samples (true positive, TP), negative samples (false positive, FP), and false negative samples (false negative, FN), respectively. IoU is defined as follows:

$$\text{IoU} = \frac{\text{area}(A_p \cap A_{gt})}{\text{area}(A_p \cup A_{gt})} \tag{13}$$

where A_p and A_{gt} denote the positions of the predicted and real frames, respectively.

The AR is defined as follows:

$$\text{AR} = \int_{0.5}^1 R(o)\text{d}o \tag{14}$$

where R denotes the recall rate and o denotes the IoU threshold. AR essentially denotes averaging over all IoUs and all categories.

To better validate the performance of the proposed method, this article introduces an AP evaluation like the Microsoft Common Objects in Context (COCO) dataset. Specifically, AP_{50} and AP_{75} are calculated according to the thresholds of 0.5 and 0.75, respectively. Meanwhile, AP is obtained by averaging ten IoU thresholds (from 0.5 to 0.95) and tends to reward models that perform better at precise locations. In addition to AP, AP_{50}, and AP_{75}, we also computed AP and AR for small, medium, and large targets, named AP_S, AP_M, AP_L, AR_S, AR_M, and AR_L, respectively.

C. Experimental Setting

In this article, the PyTorch framework is used to implement the proposed method. The size of the MR-HRSID input image is set to 800 × 800 pixels, and the size of the R-SSDD input image is set to 512 × 512. The weights for the ResNet50 backbone are initialized using a pretrained model on the ImageNet dataset while the other weights are initialized at the default settings. The loss function is optimized using Adam algorithm with an initial learning rate of 0.0001. The number of epochs is 100, the batch size of MR-HRSID is set to 2, and the batch size of R-SSDD is set to 4. The optimization is performed on a computer with an Intel i7-9700k processor, and a GeForce GTX 2070 GPU graphics card performed with Windows 10, CUDA10.0, and cuDNN as 7.6.4 as the operating system.

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TABLE I
LGCFM HYPERPARAMETER VALIDATION ON THE MR-HRSID DATASET

[1, 2, 3, ∞]	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S
Baseline	17.79	56.96	3.84	11.29	15.27	29.57	20.71
[1, 2, 3, 4]	18.2(+0.41)	57.66(+0.70)	4.60(+0.76)	11.44(+0.15)	16.38(+1.11)	29.48(-0.09)	21.01(+0.40)
[1, 2, 3, 5]	17.45(-0.34)	55.49(-1.47)	4.06(+0.22)	10.93(-0.36)	15.85(+0.58)	28.48(-1.09)	20.28(-0.43)
[1, 2, 3, 6]	18.2(+0.41)	58.25(+1.29)	3.45(-0.39)	12.09(+0.80)	15.2(-0.07)	29.3(-0.27)	21.44(+0.73)
[1, 2, 3, 7]	18.69(0.90)	59.66(+2.70)	4.47(-0.63)	12.36(+1.07)	16.56(+1.29)	29.33(-0.24)	22.23(+1.52)
[1, 2, 3, 8]	18.26(+0.47)	58.86(+1.90)	3.79(-0.05)	12.34(+1.05)	16.57(+1.30)	28.48(-1.09)	22.00(+1.29)

TABLE II
LGCFM HYPERPARAMETER VALIDATION ON THE R-SSDD DATASET

[1, 2, 3, ∞]	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S
Baseline	28.74	81	8.18	23.62	30.41	32.72	32.55
[1, 2, 3, 4]	32.02(+3.28)	83.14(+2.14)	13.68(+5.50)	27.87(+4.25)	32.1(+1.69)	36.01(+3.29)	34.84(+2.29)
[1, 2, 3, 5]	29.47(+0.73)	80.49(-0.51)	11.14(+2.96)	26.66(+3.04)	29.92(-0.49)	32.86(+0.14)	33.35(+0.80)
[1, 2, 3, 6]	30.37(+1.63)	82.44(+1.44)	10.20(+2.02)	26.91(+3.29)	30.08(-0.33)	34.52(+1.8)	35.21(+2.66)
[1, 2, 3, 7]	31.14(+2.4)	82.50(+1.5)	11.18(+3.00)	26.25(+2.63)	32.74(+2.33)	34.96(+2.24)	34.68(+2.13)
[1, 2, 3, 8]	29.89(+1.15)	81.68(+0.68)	10.06(+1.88)	25.98(+2.37)	30.18(-0.23)	33.94(+1.22)	34.84(+2.29)

D. Hyperparametric Validation and Ablation Experiments

1) STQM and LGCFM Hyperparameter Validation: In LGCFM, the kernel size of the average pooling layer we set to [1, 2, 3, 6], which enables us to obtain target features under different receptive fields. However, there are many different choices of kernel sizes for the average pooling layer, and different kernel sizes have different detection performances. Therefore, different sizes of kernels are needed to test to justify the kernel sizes set in this article on MR-HRSID and R-SSDD, respectively, as shown in Tables I and II.

The MR-HRSID and R-SSDD datasets contain many small targets, and the main role of LGCFM is to improve the detection performance of small targets, so the three metrics AP_{50}, AP_S, and AR_S are focused on. As can be seen from Tables I and II, the three metrics AP_{50}, AP_S, and AR_S in the MR-HRSID dataset are [1, 2, 3, 7], [1, 2, 3, 8], [1, 2, 3, 6], [1, 2, 3, 4], and [1, 2, 3, 5] in the descending order. In the R-SSDD dataset, AP_{50} are [1, 2, 3, 4], [1, 2, 3, 7], [1, 2, 3, 6], [1, 2, 3, 8], and [1, 2, 3, 5] in the descending order; AP_S are [1, 2, 3, 4], [1, 2, 3, 6], [1, 2, 3, 5], [1, 2, 3, 7], and [1, 2, 3, 8] in the descending order; and AR_S are [1, 2, 3, 6], [1, 2, 3, 4], [1, 2, 3, 8], [1, 2, 3, 7], and [1, 2, 3, 5] in the descending order. From the results of the comparison, the detection results of different kernel sizes are quite different and the performance on different datasets is not the same. However, from the comparison results, it is easily found that among the three metrics AP_{50}, AP_S, and AR_S, the detection results of kernel sizes [1, 2, 3, 6] and [1, 2, 3, 7] are relatively superior and more stable.

After [1, 2, 3, 6] and [1, 2, 3, 7] are chosen as the kernel sizes for the average pooling layer, the STQM is added to the network to verify the detection performance of both kernel sizes in the presence of the STQM. As can be seen in Tables III and IV, there is a large difference in the detection results of the hyperparameters [1, 2, 3, 6] and [1, 2, 3, 7] after adding the STQM. The MR-HRSID dataset with hyperparameters of [1, 2, 3, 6], the AP, AP_{50}, AP_S, AP_M, AP_L, and AR_S metrics improved by 0.96%, 1.95%, 0.16%, 1.28%, 0.47%, and 1.28%, respectively. However, the MR-HRSID with hyperparameters of [1, 2, 3, 7], the AP, AP_{50}, AP_S, AP_M, AP_L, and AR_S metrics decreased by 0.3%, 1.08%, 0.31%, 0.26%, 0.8%, 0.31%, and 0.25%, respectively, compared to the baseline. In the R-SSDD dataset, the detection performance of the two hyperparameters is varied. In hyperparameters [1, 2, 3, 6], AP_{50} improves by 2.24% and the metrics AP_S and AP_{50}, which are specialized for evaluating small targets, improve by 3.36% and 2.34%, respectively; whereas in hyperparameters [1, 2, 3, 7], AP_{50}, AP_S, and AR_S improve by 0.76%, 2.35%, and 0.43%, respectively, whose performance is far below that of the hyperparameters [1, 2, 3, 6]. Summarizing the results of the above analysis, the detection performance of the network is better and robust in the network with hyperparameters [1, 2, 3, 6].

We analyze the reasons for the performance difference of LGCFM under different hyperparameters from the perspective of the target scale distribution of the two datasets. As shown in Fig. 9, the target scale differences in both MR-HRSID and R-SSDD datasets are relatively large, and the number of large-scale targets and small-scale targets is greater (in this article, the target area of less than 300 pixels is classified as relatively small as a small target, the target area of more than 300 pixels but less than 600 pixels is classified as a medium-scale target, and the target area of larger than 600 pixels are large-scale targets). On the one hand, among these three scales of targets, small targets contain less information and are the most difficult

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TABLE III
STQM AND LGCFM HYPERPARAMETER VALIDATION ON MR-HRSID DATASET

[1, 2, 3, ∞]	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S
Baseline	17.79	56.96	3.84	11.29	15.27	29.57	20.71
STQM+LGCFM ([1, 2, 3, 6])	18.75 **(+0.96)**	58.91 **(+1.95)**	4.00 **(+0.16)**	11.95 **(+0.66)**	16.55 **(+1.28)**	30.04 **(+0.47)**	21.99 **(+1.28)**
STQM+LGCFM ([1, 2, 3, 7])	17.49 (-0.30)	55.88 (-1.08)	3.53 (-0.31)	11.03 (-0.26)	14.47 (-0.80)	29.26 (-0.31)	20.46 (-0.25)

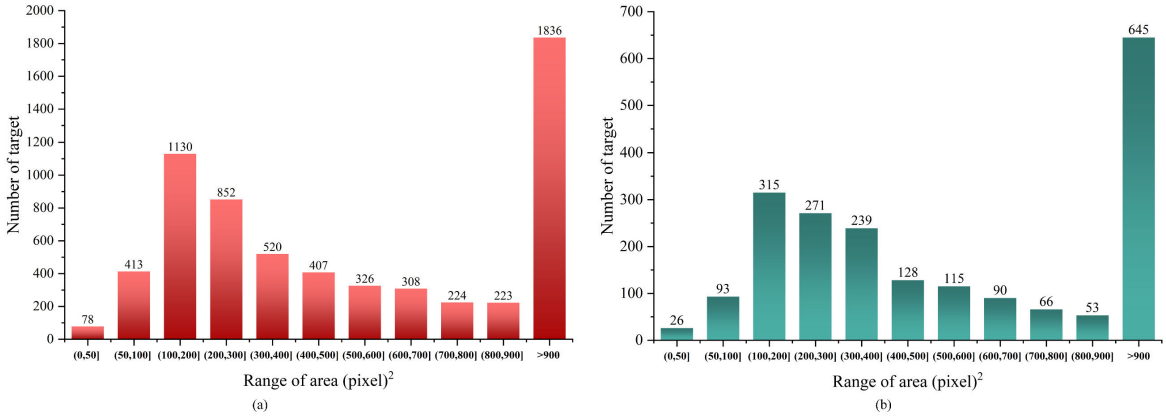


Fig. 9. Target scale statistics: (a) MR-HRSID and (b) R-SSDD.

to detect. When the kernel of the average pooling layer in the LGCFM is set to [1, 2, 3, 7], the size of the receptive field it obtains is much larger than the scale of the small target. Thus, it will capture more noise around the small target, which will be not favorable to the classification and regression of the small target and make the network's ability to detect the small target decrease. For example, in the MR-HRSID dataset, each metric decreases when using kernels of [1, 2, 3, 7]. In addition, in the R-SSDD dataset, AP_S and AR_S are improved by 3.36% and 2.34%, respectively, using the kernel of [1, 2, 3, 6], which is greater than using the kernel of [1, 2, 3, 7]. On the other hand, although setting the kernel of the average pooling layer in LGCFM to [1, 2, 3, 7] facilitates the detection of large targets, the large targets contain rich information, and the detection results of their large targets are still better even without using LGCFM. For example, in the baseline without LGCFM, the two datasets, HRSID and R-SSDD, have an AP_L of 29.57% and 32.72%, respectively, which is the best among the three scales. Therefore, combining the above analysis and the scale distribution of targets in SAR images, the [1, 2, 3, 6] is selected as the hyperparameters of the LGCM average pooling layer.

After determining the hyperparameters of LGCFM, the positions of the two modules (STQM and LGCFM) are explored to obtain the best detection performance. As can be seen in Tables V and VI, the detection results of (STQM + LGCFM) are better than (LGCFM + STQM), both on the MR-HRSID dataset and on the R-SSDD dataset. The reason for the difference in detection results is that in the shallow feature C_3, although it contains more information about small targets, which is suitable for small target localization and classification, it also contains more noise. In the combination of (LGCFM + STQM), LGCFM is first utilized to capture the local and global contextual information of the target, but since C_3 contains more noise, LGCFM inevitably captures a large amount of noise; when STQM is utilized to process the output of LGCFM, STQM is unable to fully process all the noise in the output feature maps of LGCFM, thus making the combined approach of (LGCFM + STQM) a limited enhancement in both datasets. In the (STQM + LGCFM) combination, the noise in C_3 is first filtered by using STQM, and then, the local and global context information in C_3' is learned using LGCFM. After the suppression of noise in C_3 by STQM, C_3' is input to LGCFM, and pixels within the kernel size are replaced by a mean value, further weakening the effect of noise on the target, thus making both datasets more significant in the STQM + LGCFM combination.

2) Module Ablation Study: After testing the hyperparameters of the LGCFM and the positions of the two modules, ablation experiments are performed on individual modules to examine the role of different modules in the network, as shown in Tables VII and VIII.

1) *STQM*: In the MR-HRSID dataset, after adding STQM, AP_{50} improves from 56.96% to 58% in the baseline, AP_S decreases by 0.06%, and AR_S improves by 0.28%; in the R-SSDD dataset, AP_{50} improves from 81% to 83% in the baseline, AP_S improves by 2.36%, and AR_S improves by 0.91%. In both datasets, STQM has limited enhancement for small targets. For medium- and large-scale targets, AP_M, AP_L, AR_M, and AR_L are boosted by 1.39%, 1.12%, 1.27%, and 1.68% in the MR-HRSID dataset, respectively, and these four metrics are boosted by 1.27%, 2.67%, 1.72%, and 3.02% in R-SSDD, respectively. In terms of the four metrics,

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TABLE IV
STQM AND LGCFM HYPERPARAMETER VALIDATION ON THE R-SSDD DATASET

[1, 2, 3, ∞]	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S
Baseline	28.74	81	8.18	23.62	30.41	32.72	32.55
STQM+LGCFM ([1, 2, 3, 6])	31.33 (+2.59)	83.24 **(+2.24)**	12.74 (+4.56)	26.98 **(+3.36)**	31.95 (+1.54)	35.24 (+2.52)	34.89 **(+2.34)**
STQM+LGCFM ([1, 2, 3, 7])	31.78 **(+3.04)**	81.76 (+0.76)	14.62 **(+6.44)**	25.97 (+2.35)	32.73 **(+2.32)**	36.46 **(+3.74)**	32.98 (+0.43)

TABLE V
STQM AND LGCFM POSITION VERIFICATION ON THE MR-HRSID DATASET

position	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S
Baseline	17.79	56.96	3.84	11.29	15.27	29.57	20.71
STQM+LGCFM	18.75 **(+0.96)**	58.91 **(+1.95)**	4.00 **(+0.16)**	11.95 **(+0.66)**	16.55 (+1.28)	30.04 **(+0.47)**	21.99 **(+1.28)**
LGCFM+STQM	18.32 (+0.53)	57.43 (+0.47)	3.97 (+0.13)	11.29 (+0.00)	17.27 **(+2.00)**	29.92 (+0.35)	20.69 (-0.02)

TABLE VI
STQM AND LGCFM POSITION VERIFICATION ON THE R-SSDD DATASET

position	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S
Baseline	28.74	81	8.18	23.62	30.41	32.72	32.55
STQM+LGCFM	31.33 **(+2.59)**	83.24 **(+2.24)**	12.74 **(+4.56)**	26.98 **(+3.36)**	31.95 **(+1.54)**	35.24 **(+2.52)**	34.89 **(+2.34)**
LGCFM+STQM	30.34 (+1.60)	82.17 (+1.17)	11.58 (+3.40)	26.9 (+3.28)	30.6 (+0.19)	34.16 (+1.44)	34.31 (+1.76)

TABLE VII
ABLATION EXPERIMENTS ON THE MR-HRSID DATASET

Baseline	STQM	LGCFM	Center Loss	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S	AR_M	AR_L
√	-	-	-	17.79	56.96	3.84	11.29	15.27	29.57	20.71	27.37	39.59
√	√	-	-	18.55	58	4.21	11.23	16.66	30.69	20.99	**28.64**	**41.27**
√	-	√	-	18.2	58.25	3.45	12.09	15.2	29.3	21.44	27.37	39.31
√	√	√	-	18.75	58.91	4.00	11.95	16.55	30.04	21.99	28.37	39.90
√	√	√	√	**19.11**	**60.77**	**4.27**	**12.61**	**17.03**	**30.22**	**22.24**	28.37	39.87

TABLE VIII
ABLATION EXPERIMENTS ON THE R-SSDD DATASET

Baseline	STQM	LGCFM	Center Loss	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S	AR_M	AR_L
√	-	-	-	28.74	81	8.18	23.62	30.41	32.72	32.55	39.1	44.12
√	√	-	-	30.86	83	11.71	26.26	31.68	35.39	33.46	**40.82**	**47.14**
√	-	√	-	30.37	82.44	10.2	26.91	30.08	34.52	**35.21**	37.21	45.55
√	√	√	-	31.33	83.24	12.74	26.98	**31.95**	35.24	34.89	39.51	45.63
√	√	√	√	**31.92**	**84.31**	**13.35**	**28.48**	31.32	**36.14**	35.16	38.61	46.3

AP_M, AP_L, AR_M, and AR_L, STQM is more favorable for medium- and large-scale targets.

2) *LGCFM*: With the addition of LGCFM, AP_S and AR_S are improved by 0.8% and 0.93% in the MR-HRSID dataset, respectively; AP_S and AR_S are improved by 2.29% and 2.66% in R-SSDD, respectively. For medium- and large-scale targets, the enhancement is very small in this module. As elaborated in Section II-C, contextual information helps in the detection of small targets.
3) *STQM + LGCFM*: When STQM and LGCFM together are embedded into the network, compared to baseline, AP, AP_{50}, AP_{75}, AP_S, AP_M, AP_L, and AR_S in the

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TABLE IX
DETECTION RESULTS OF DIFFERENT ORIENTATION DETECTORS ON MR-HRSID

Methods	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S	AR_M	AR_L
Rotated Reppoints	10.99	37.58	3.28	5.26	8.74	23.05	11.17	23.2	35.44
R^3Det	12.4	38.83	3.75	3.43	13.66	27.09	7.74	21.51	34.02
YOLOv5-obb	5.75	22.49	0.57	1.64	9.6	11.69	6.31	19.92	27.09
Rotated-ATSS	12.67	39.15	3.67	4.8	14	26.27	9.25	23.36	34.18
Rotated Faster R-CNN	11.04	38.6	1.89	5.61	13.45	19.17	12.73	24.22	27.06
CSL	6.74	24.04	1.66	2.44	10.28	12.76	6.93	16.21	18.39
Gliding Vertex	17.3	57.75	4.08	11.02	17.02	27.6	18.33	27.09	36.21
Proposed	**19.11**	**60.77**	**4.27**	**12.61**	**17.03**	**30.22**	**22.24**	**28.37**	**39.87**

MR-HRSID dataset improved by 0.96%, 1.96%, 0.16%, 0.66%, 1.28%, 0.47%, and 1.28%, respectively; in R-SSDD, these seven metrics are increased by 2.59%, 2.24%, 4.56%, 3.36%, 1.54%, 2.52%, and 2.34%, respectively. In summary, the overall improvement of the detection performance of the network is larger. Especially in R-SSDD, the AP_S and AR_S of small targets are improved by 3.36% and 2.34%, respectively.

4) *STQM + LGCFM + Center Loss*: To further constrain the center coordinates and shapes of the oriented bounding box to balance the loss contributions among different variables, the center loss is added to the network, which further extends the network performance improvement in terms of various metrics. Compared to baseline, AP, AP_{50}, AP_{75}, AP_S, AP_M, AP_L, and AR_S in the MR-HRSID dataset are improved by 1.32%, 3.81%, 0.43%, 1.32%, 1.76%, 0.65%, and 1.53%, respectively; and these seven metrics in R-SSDD are improved by 3.18%, 3.31%, 5.17%, 4.86%, 0.91%, 3.42%, and 2.61%, respectively. Combining the results of the above analysis, the three modules proposed in this article have different degrees of enhancement for the three scales of objectives.

E. Detection Results of Different Oriented Target Detection Networks

In Section III-D, we verify the detection performance of each component in the proposed network. To further test the advancement of the proposed network, it is necessary to compare it with the state-of-the-art oriented target networks in the field of remote sensing. In this article, the networks we choose are Rotated Reppoints [46], R^3Det [47], YOLOv5-obb [48], Rotated-ATSS [49], Rotated Faster R-CNN [50], CSL [51], and Gliding Vertex [52]. We chose these networks because they are advanced oriented target detection networks in remote sensing and are widely used for the objects being compared.

As shown in Table IX, we count the detection results of eight networks. From Table IX, we can see that the method proposed in this article achieves 60.77% AP_{50} on MR-HRSID, which is 3.02% higher than Gliding Vertex, which has the highest AP_{50} among the compared networks, which fully demonstrates the superiority of the proposed network. In addition, for the detection results of targets at different scales, the AP_s and AR_s of the proposed network in this article are 12.61% and 22.24%, which are 1.59% and 3.91% higher than Gliding Vertex, which also has the highest AP_S and AR_S in the compared network, respectively; the AP_M and AR_M are 0.01% and 1.28% higher than Gliding Vertex, respectively; and AP_L and AR_L are 2.62% and 3.66% higher than Gliding Vertex, respectively. From the above metrics, the detection performance of the proposed network is optimal.

Figs. 10 and 11 show the detection results of eight oriented detection networks on MR-HRSID. Fig. 10 shows the detection results within an inland river. From Fig. 10, it can be seen that the scene contains a lot of land background and the scale of the ships within the scene is very small, which is a great challenge for the neural network. From the detection results, the detection performance of the five networks, Rotated Reppoints, R^3Det, YOLOv5-obb, Rotated-ATSS, and CSL, decreases dramatically in this complex scene. For Rotated Faster R-CNN detected some of the ships, but at the same time, it produced many missed detections and false alarms. In this scenario, the best detection results are in Gliding Vertex and the network proposed. Looking closely, the number of missed detections and false alarms of ships is more in Gliding Vertex compared to the proposed network. Fig. 11 shows a harbor scene in which ships of different scales are docked. From the detection results, Rotated Reppoints, R^3Det, YOLOv5-obb, Rotated-ATSS, and CSL miss many ships, and Rotated Faster R-CNN detects ships but also generates the most false alarms among all networks. Similarly, Gliding Vertex generates many false alarms. For the proposed network, the greatest number of ships are detected, and the number of false alarms is the least compared to Rotated Faster R-CNN and Gliding Vertex.

Table X shows the detection results of different networks on R-SSDD. For AP and AP_{50}, the proposed network outperforms the optimal R^3Det and Rotated-ATSS among the compared networks by 2.16% and 9.44%, respectively; while for the high-precision localization metrics AP_{75}, the R^3Det is 16.13%, which is the highest among all networks. For AP_S and AR_S, Rotated Faster R-CNN is 33.66% and 43.09%, respectively, the highest among all networks. This is because of the region proposal network (RPN) structure in Rotated Faster R-CNN, the target is filtered by RPN, which can effectively filter the

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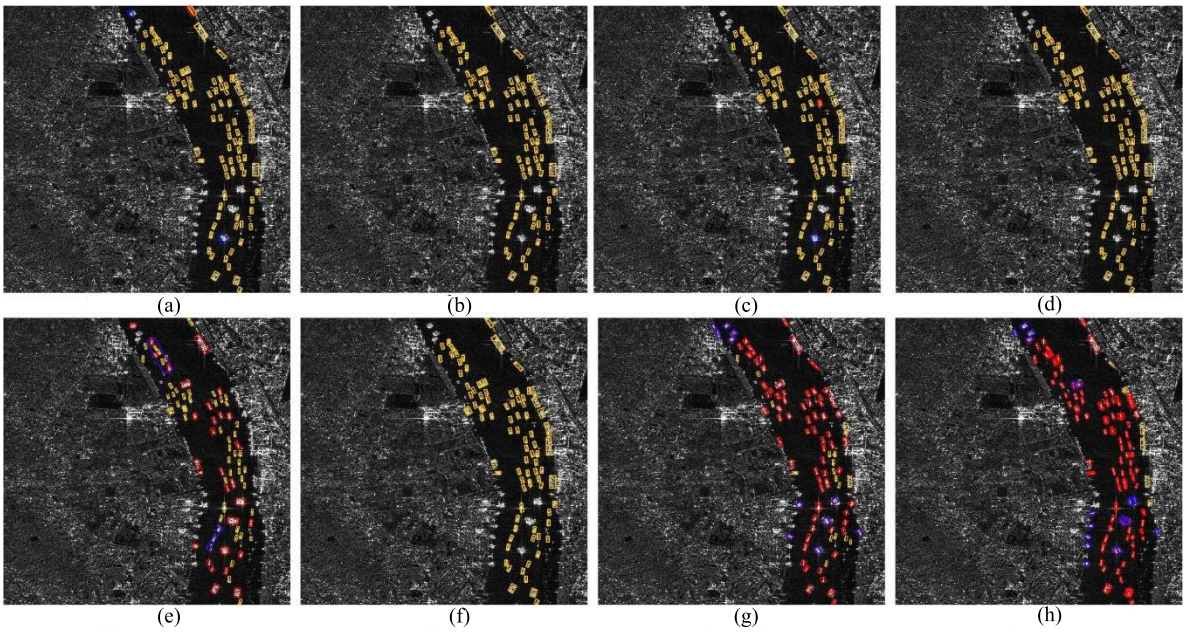


Fig. 10. Inland waterways detection results for the eight networks in the MR-HRSID dataset. Red boxes indicate correctly predicted ships, orange boxes indicate missed detections, and blue boxes indicate false alarms: (a) Rotated Reppoints; (b) R^3Det; (c) YOLOv5-obb; (d) Rotated-ATSS; (e) Rotated Faster R-CNN; (f) CSL; (g) Gliding Vertex; and (h) Proposed.

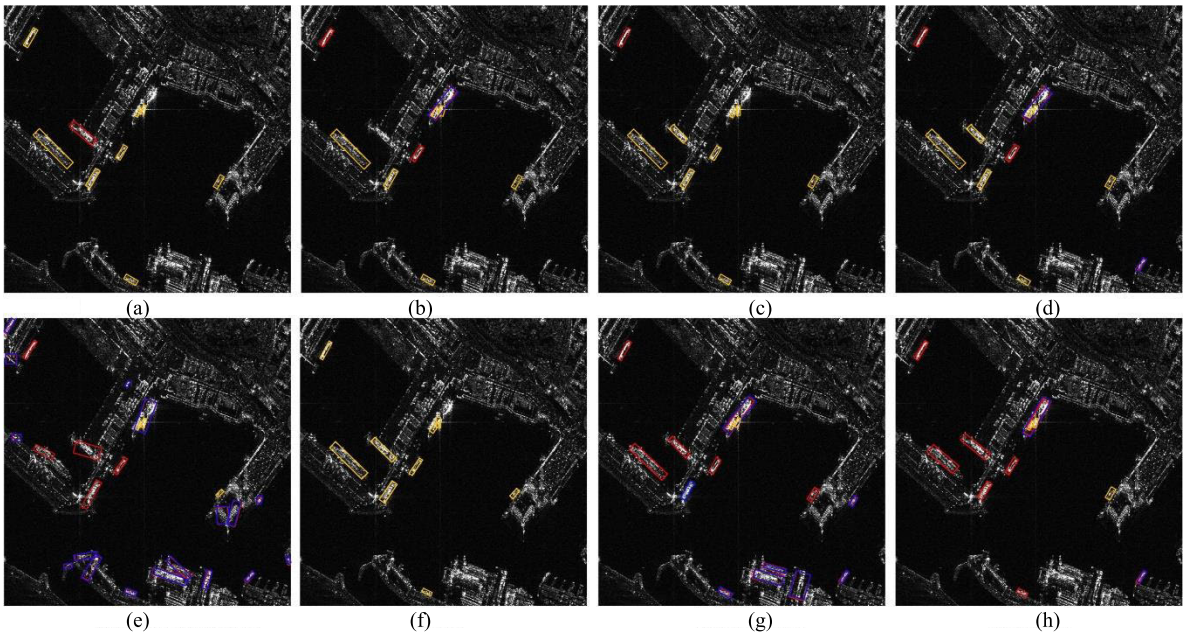


Fig. 11. Inshore detection results for the eight networks in the MR-HRSID dataset. Red boxes indicate correctly predicted ships, orange boxes indicate missed detections, and blue boxes indicate false alarms: (a) Rotated Reppoints; (b) R^3Det; (c) YOLOv5-obb; (d) Rotated-ATSS; (e) Rotated Faster R-CNN; (f) CSL; (g) Gliding Vertex; and (h) Proposed.

noise in the target and keep only the candidate target so that the fully connected layer in the network only needs to focus on the candidate target, which can effectively improve the precision and recall. The highest AP_M and AR_M is for R^3Det, which indicates that for medium-scale targets in R-SSDD, R^3Det can capture more information and thus improve the precision and recall of such targets. For AP_L and AR_L, the proposed network occupies a significant advantage of 36.14% and 46.3%, which are 5.18% and 9.24% higher than the optimal R^3Det and Rotated Reppoints in the compared networks, respectively. Combined with the above analysis, the detection performance of different networks for different scales of targets in R-SSDD is different, but we combine all the metrics to see that the proposed network is superior to the networks being compared in this article.

Figs. 12 and 13 show the detection results of the eight networks in the R-SSDD dataset. From the detection results, five networks, namely, Rotated Reppoints, R^3Det, YOLOv5-obb, Rotated-ATSS, and CSL, appear to be powerless for the docked ships in Fig. 12. Rotated Faster R-CNN detected

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TABLE X
RESULTS OF DIFFERENT ORIENTATION DETECTORS ON R-SSDD

Methods	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AR_S	AR_M	AR_L
Rotated Reppoints	28.47	70.7	14.06	33.54	30.47	23.75	41.44	38.44	37.06
R^{3}Det	29.76	74.47	**16.13**	27.37	**33.55**	29.96	33.46	**38.93**	36.3
YOLOv5-obb	13	46.65	0.85	16.48	16.27	8.9	29.31	24.34	18.53
Rotated-ATSS	29.46	74.87	15.83	28.56	32.23	29.17	34.73	36.89	35.13
Rotated Faster R-CNN	26.28	69.48	10.92	**33.66**	29.08	19.6	**43.09**	35.25	28.78
CSL	17.04	55.12	4.58	21.57	19.54	14.43	29.52	24.43	19.03
Gliding Vertex	6.71	39.29	0.06	10.92	7.06	4.14	18.83	10.25	7.82
Proposed	**31.92**	**84.31**	13.35	28.48	31.32	**36.14**	35.16	38.61	**46.3**

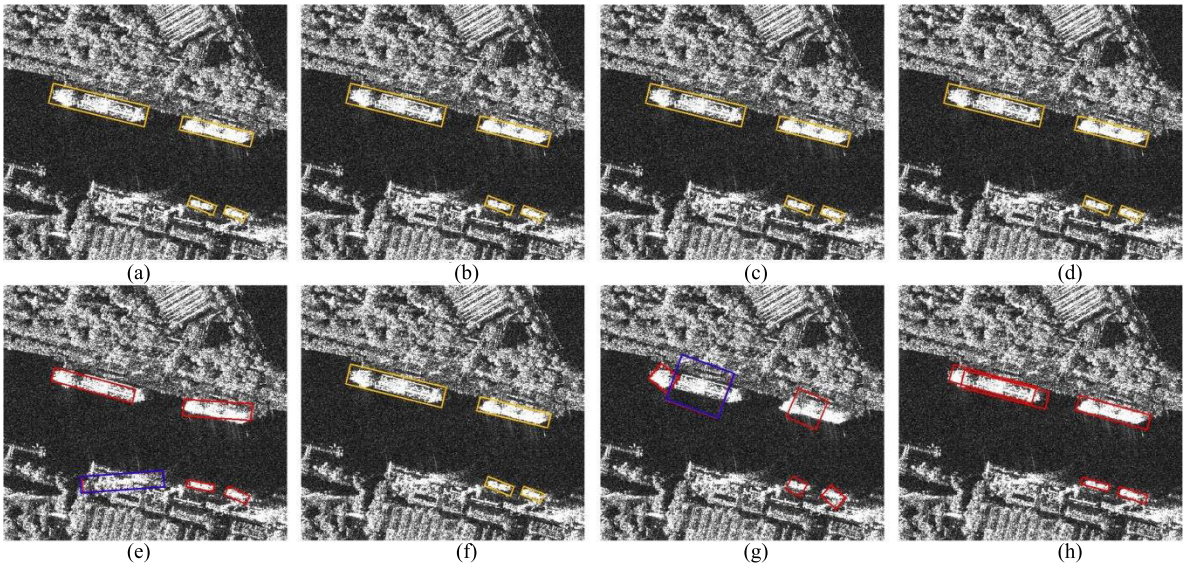


Fig. 12. Inshore detection results of the eight networks in the R-SSDD dataset. Red boxes indicate correctly predicted ships, orange boxes indicate missed detections, and blue boxes indicate false alarms: (a) Rotated Reppoints; (b) R^{3}Det; (c) YOLOv5-obb; (d) Rotated-ATSS; (e) Rotated Faster R-CNN; (f) CSL; (g) Gliding Vertex; and (h) Proposed.

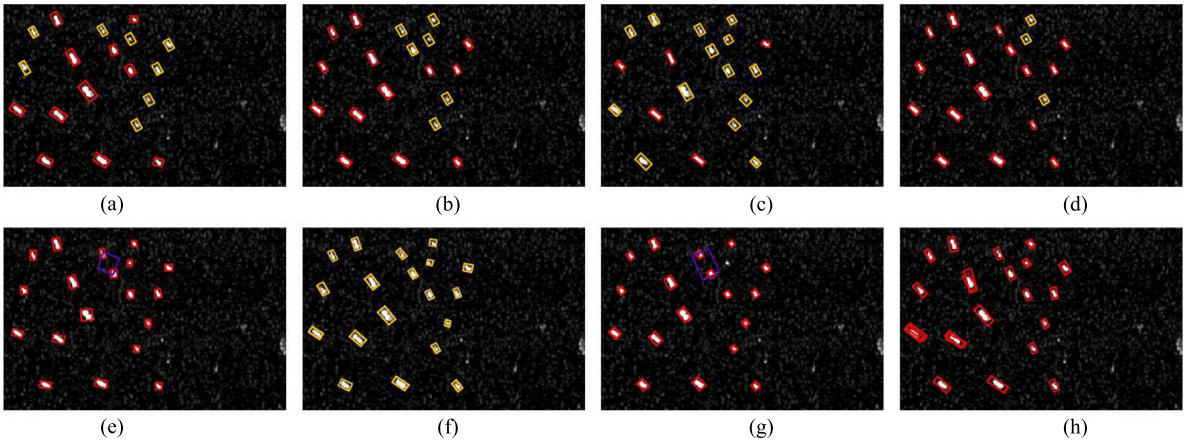


Fig. 13. Offshore detection results of the eight networks in the R-SSDD dataset. Red boxes indicate correctly predicted ships, orange boxes indicate missed detections, and blue boxes indicate false alarms: (a) Rotated Reppoints; (b) R^{3}Det; (c) YOLOv5-obb; (d) Rotated-ATSS; (e) Rotated Faster R-CNN; (f) CSL; (g) Gliding Vertex; and (h) Proposed.

all the ships but generated a false alarm. Gliding Vertex detected three ships, again producing a false alarm. For the proposed network, all ships were detected in this scenario with no false alarms, and it has the best detection performance. Fig. 13 shows small targets in the R-SSDD dataset. Rotated Reppoints, R^{3}Det, YOLOv5-obb, and Rotated-ATSS have

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detected detections but the number of missed detections by the ships is also high. Rotated Faster R-CNN, Gliding Vertex, and the proposed network detected all the detections in the scene; however, Rotated Faster R-CNN and Gliding Vertex each had a false alarm.

Combining the nine-evaluation metrics and the visualization results of the two datasets, we can conclude that the proposed network shows excellent detection performance both for small-scale ships in inland rivers and for inshore ships docked in ports; and its performance is optimal among all the networks in this article on different datasets, which also shows its better robustness.

IV. Conclusion

In this article, a novel algorithm for ship orientation detection in SAR images is presented. The proposed method encompasses several key components to enhance the overall performance of ship detection. First, an STQM aimed at mitigating the impact of background noise in SAR images during network learning is designed. This module effectively reduces noise interference. Second, an LGCFM that integrates local and global contextual information from SAR images is constructed. This module significantly improves the detection accuracy of small targets. Additionally, the center loss technique to mitigate the sensitivity of the target to the GT shape is incorporated.

In comparison to the CNN method described in this article, our proposed network demonstrates superior performance in ship detection accuracy, false alarm rate, and localization accuracy. Conducting experiments on the MR-HRSID and R-SSDD datasets, the results validate the effectiveness of our proposed network.

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